<u>Association Mining Exercice</u>

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Step 1 : Compute L(1)

L(1) = {Milk, Beer, Diapers, Bread, Butter, Cookies}

All 1-itemsets are interesting

ItemSet	Count
Milk	5
Beer	4
Diapers	7
Bread	5
Butter	4
Cookies	4

Step 2 : Compute L(2)

L(2) = {(Milk, Beer), (Milk, Diapers),(Milk, Cookies),(Bread, Butter),(Bread, Milk),(Bread, Diapers),(Butter, Milk),(Butter, Diapers),(Cookies, Diapers),(Cookies, Beer)}

Only (Milk, Diapers) and (Bread, Butter) are interesting.

ItemSet	Count
(Milk, Beer)	1
Milk, Diapers)	4
(Milk, Cookies)	1
(Bread, Butter)	5
(Bread, Milk)	3
(Bread, Diapers)	3
(Butter, Milk)	3
(Butter, Diapers)	3
(Cookies,Diapers)	2
(Cookies, Beer)	2

Step 3 : Compute L(3)

L(3) = {(Milk, Diapers, Bread), (Milk, Diapers, Butter), (Milk, Bread, Butter), (Bread, Butter, Diapers), (Milk, Diapers, Cookies), (Beer, Cookies, Diapers)}

There is no 3-itemsets that are interesting

ItemSet	Count
(Milk, Diapers, Bread)	2
(Milk, Diapers, Butter)	2
(Milk, Bread, Butter)	2
(Bread, Butter, Diapers)	2
(Milk, Diapers, Cookies)	2
(Beer, Cookies, Diapers)	2

Step 4 : Compute L(4)

L(4) = {(Milk, Diapers, Bread, Butter)}

ItemSet	Count
(Milk, Diapers, Bread, Butter)	2

<u>Confidence Calculation :</u>

2-ItemSets :

(Milk, Beer) : conf(Milk => Beer) = 1/5 and conf(Beer => Milk)= 1/4
(Milk, Diapers) : conf(Milk => Diapers) = 4/5 and conf(Diapers => Milk) = 4/7
(Milk, Cookies) : conf(Milk => Cookies) = 1/5 conf(Cookies => Milk) = 1/4
(Bread, Butter) : conf(Bread => Butter) = 5/5 conf(Butter => Bread) = 5/4
(Bread, Milk) : conf(Bread => Milk) = 3/5 conf(Milk => Bread) = 3/5
(Bread, Diapers) : conf(Bread => Diapers) = 3/5 conf(Diapers => Bread) = 3/7
(Butter, Milk) : conf(Butter => Milk) = 3/4 conf(Milk => Butter) = 3/5
(Cookies, Diapers) : conf(Cookies => Diapers) = 2/4 conf(Diapers => Cookies) = 2/7
(Cookies, Beer) : conf(Cookies => Beer) = 2/4 conf(Beer => Cookies) = 2/4

3-ItemSets :
(Milk, Diapers, Bread)
conf(Milk, Diapers => Bread) = 2/4
conf(Milk, Bread => Diapers) = 2/5
conf(Diapers, Bread => Milk) = 2/7

(Milk, Diapers, Butter)
conf(Milk, Diapers => Butter) = 2/4

conf(Milk, Butter => Diapers) = 2/5
conf(Diapers, Butter => Milk) = 2/7

(Milk, Bread, Butter)
conf(Milk, Bread => Butter) = 2/5
conf(Milk, Butter => Bread) = 2/5
conf(Bread, Butter => Milk) = 2/5

(Bread, Butter, Diapers)
conf(Bread, Butter => Diapers) = 2/5
conf(Bread, Diapers => Butter) = 2/5
conf(Butter, Diapers => Bread) = 2/4

(Milk, Diapers, Cookies)
conf(Milk, Diapers => Cookies) = 2/4
conf(Milk, Cookies => Diapers) = 2/5
conf(Diapers, Cookies => Milk) = 2/7

(Beer, Cookies, Diapers)
conf(Beer, Cookies => Diapers) = 2/4
conf(Beer, Diapers => Cookies) = 2/4
conf(Cookies, Diapers => Beer) = 2/4